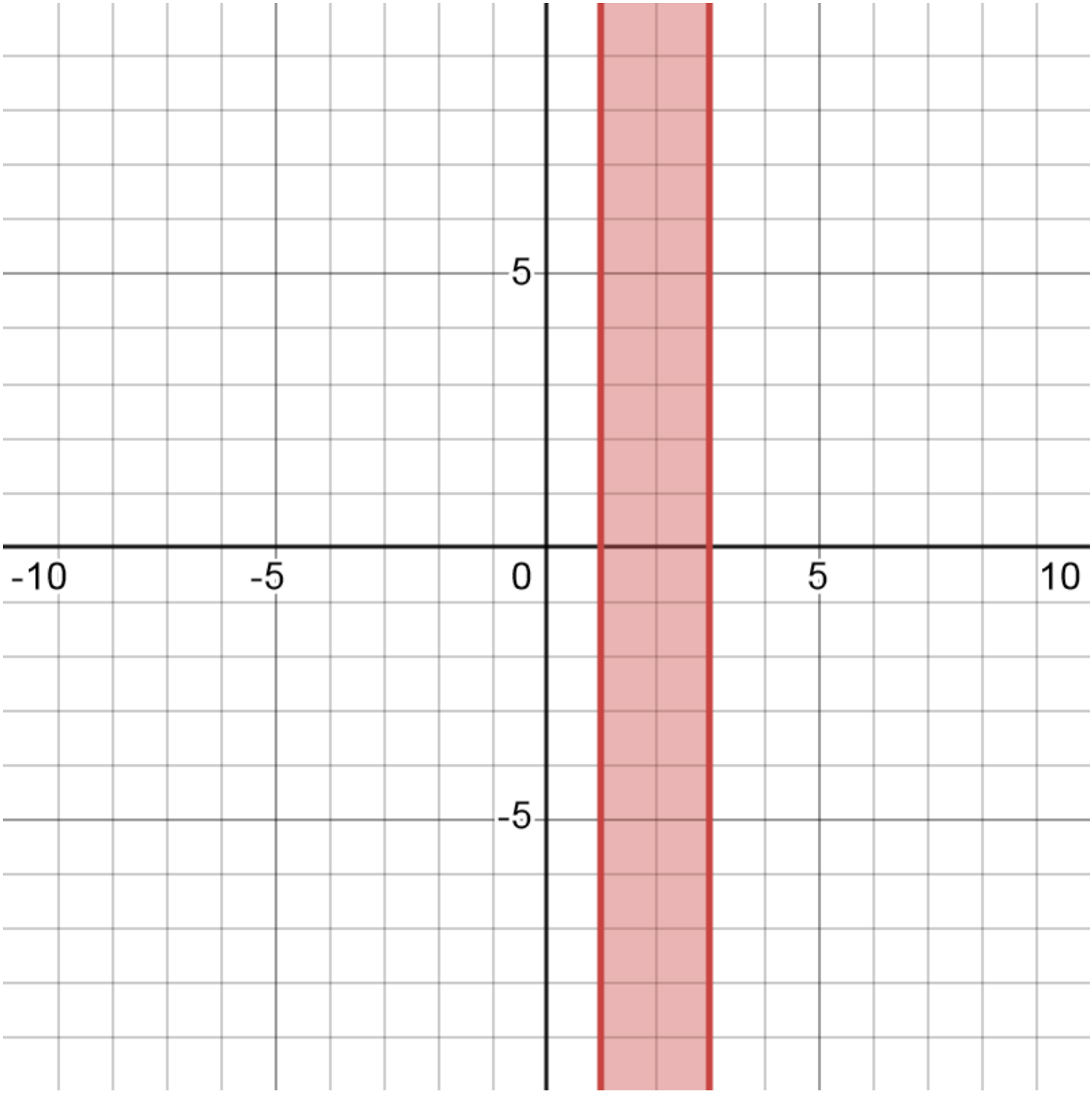
Lecture 9 (Lines in Polyhedra)

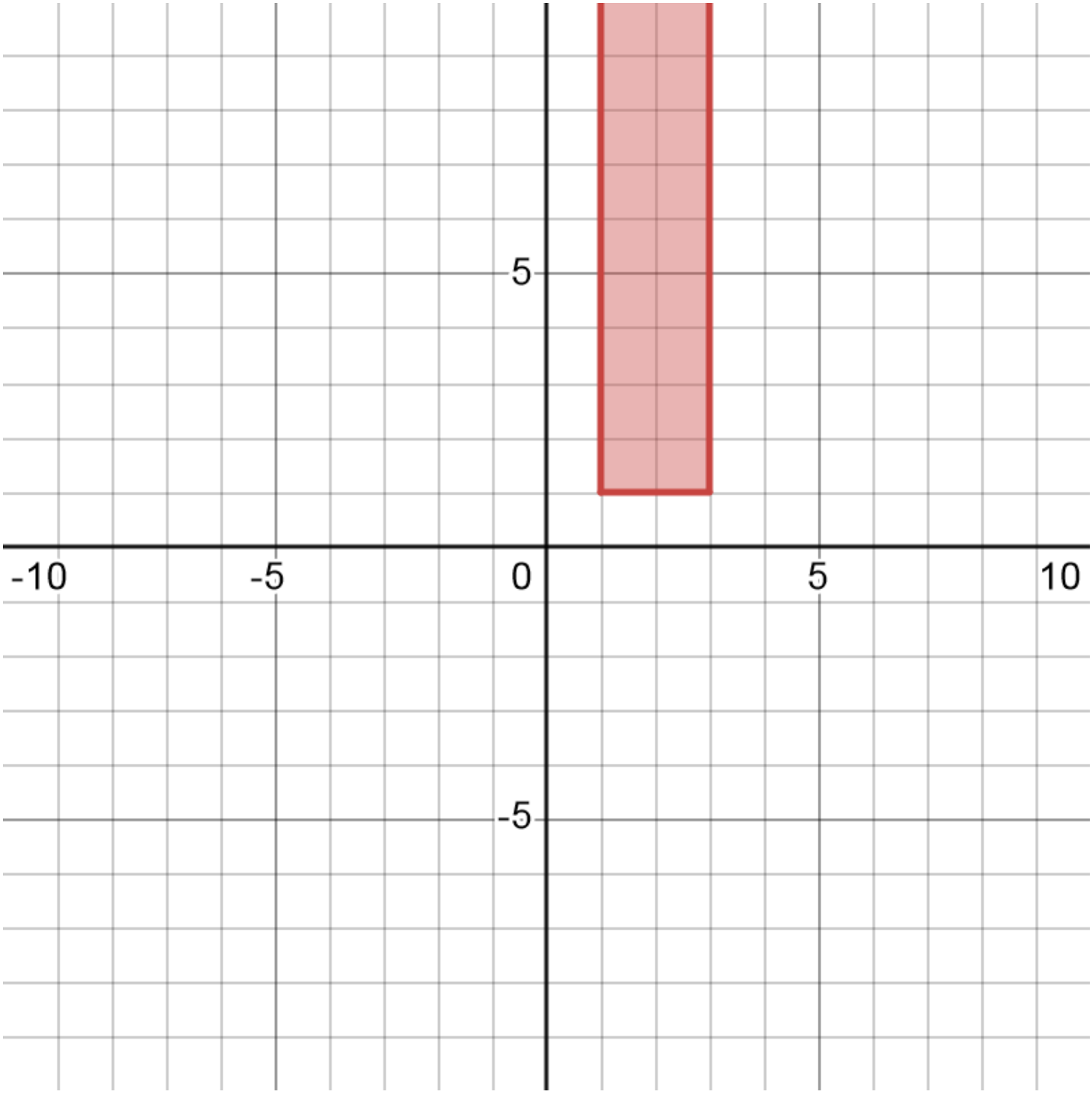
February 10, 2026

Lines in Polyhedra

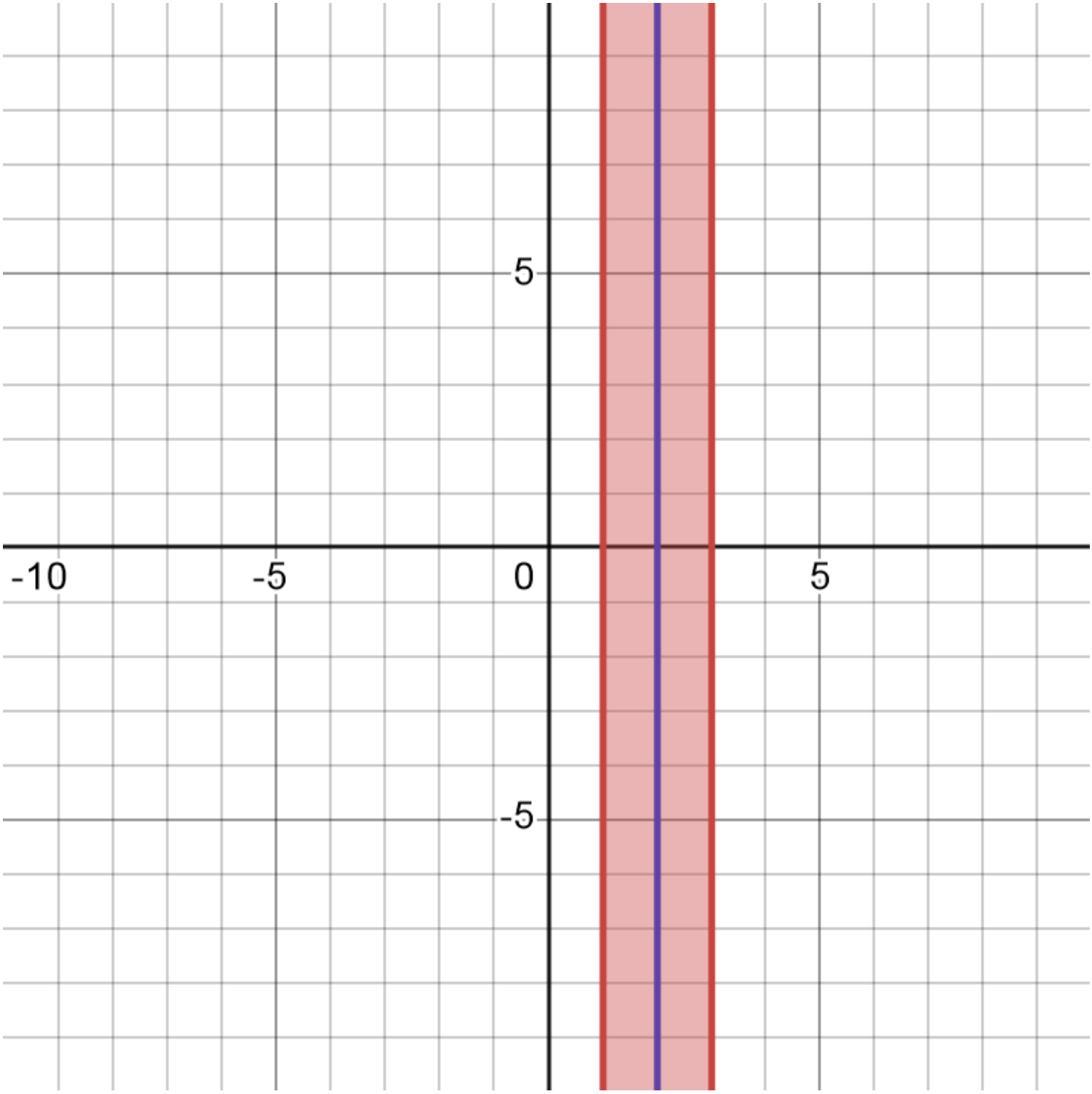
Let's start with a question. Does every (non-empty) polyhedron have an extreme point? The answer is no. Consider the following polyhedron:



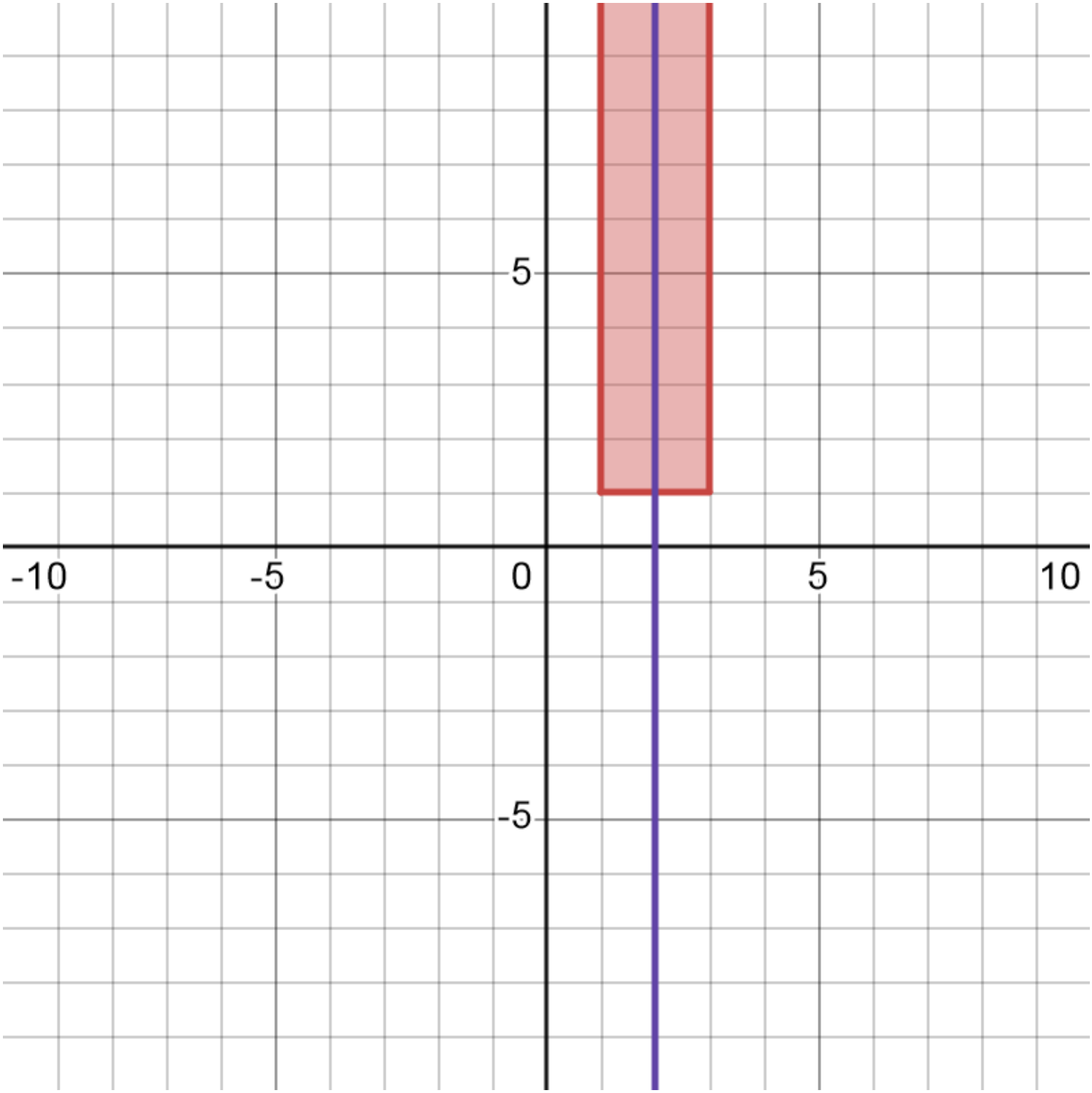
There are no corners, so there are no extreme points. If we add one additional constraint, we can get extreme points though.



It seems like any polyhedron that contains a full line doesn't contain any corners:



But a non-empty polyhedron that doesn't contain a line contains extreme points.



Let's try making this rigorous. We say a polyhedron $P \subseteq \mathbb{R}^n$ **contains a line** if $\exists \vec{x} \in P$ and $\vec{d} \in \mathbb{R}^n$ such that $\vec{d} \neq \vec{0}$ and such that

$$\vec{x} + \lambda \vec{d} \in P \; \forall \lambda \in \mathbb{R}$$

Theorem

Let $P = \{\vec{x} \in \mathbb{R}^n | \vec{a}_i \cdot \vec{x} \geq b_i, i = 1, \ldots, m\}$ be non-empty and suppose $m \geq n$. Then the following are equivalent:

1. P has at least one extreme point
2. P does not contain a line
3. There are n vectors in $\{\vec{a}_1, \ldots, \vec{a}_m\}$ which are L.I.

Proof

(2. $\implies$ 1.)

Let $\vec{x} \in P$, let $I = \left\{ i | \vec{a}_i \cdot \vec{x} = \vec{b} \right\}$, and let $S = \{\vec{a}_i | i \in I\}$. If there are n L.I. vectors in S, we are done. If we instead have a maximum of $k < n$ L.I. vectors in S, we will describe a process for adding more active constraints to S. As $k < n$, then there exists $\vec{d} \in \mathbb{R}^n$ such that $\vec{a}_i \cdot \vec{d} = 0$ for all $i \in I$. Now, let

$$\vec{y} = \vec{x} + \lambda \vec{d},\ \lambda \in \mathbb{R}$$

As there is not a line in P, then there must exist some λ^* such that for some $j \neq I$ we have

$$\vec{a}_j \cdot \left(\vec{x} + \lambda^* \vec{d} \right) = b_j$$

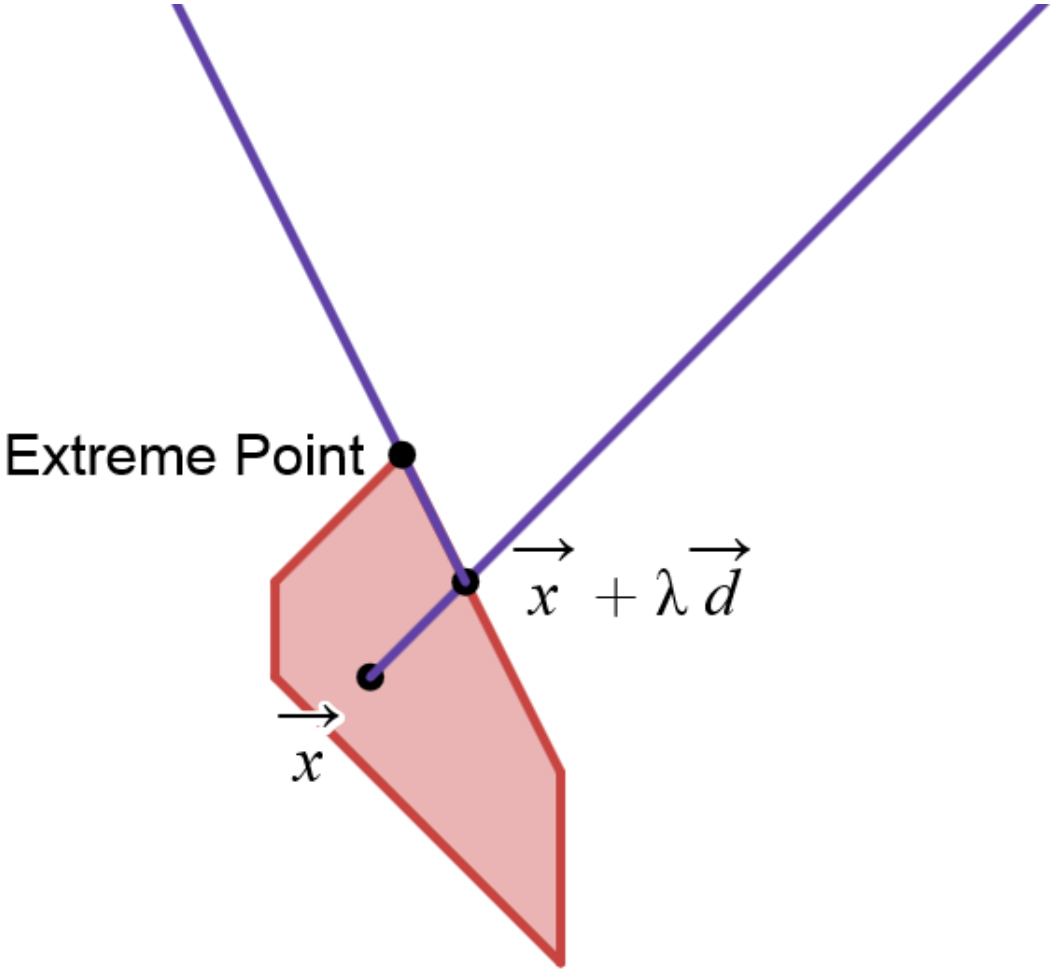
We know $\vec{a}_j$ is not a linear combination of vectors in S. To show this, suppose $\vec{a}_j$ is a linear combination of vectors in S. Then

$$\vec{a}_j \cdot \left(\vec{x} + \lambda^* \vec{d} \right) = \vec{a}_j \cdot \vec{x} + \lambda^* \vec{a}_j \cdot \vec{d} = \vec{a}_j \cdot \vec{x} = b_j$$

which would imply $j \in I$. Thus to avoid contradiction, we must have $\vec{a}_j$ not as a linear combination of vectors in S. We now can add $\vec{a}_j$ to S and repeat this process for $\vec{x} + \lambda^* \vec{d}$. As

$$\vec{a}_i \cdot \left(\vec{x} + \lambda^* \vec{d} \right) = \vec{a}_i \cdot \vec{x} + \lambda^* \vec{a}_i \cdot \vec{d} = \vec{a}_i \cdot \vec{x} = b_i\ \forall i \in S$$

then we see we now will have $k+1$ active constraints at $\vec{x} + \lambda^* \vec{d}$. Simply repeat this process until there are n L.I. vectors in S. As these will correspond to n L.I. active constraints in P, this corresponding point will be a BFS and thus an extreme point.



The geometric intuition is shown above.

(1. $\implies$ 3.)

An extreme point is a BFS. A BFS needs n L.I. active constraints. For n L.I. active constraints to exist, then there needs to exist n L.I. vectors in $\{\vec{a}_1, \ldots, \vec{a}_m\}$.

(3. $\implies$ 1.)

Suppose n of the $\vec{a}_i$ are L.I. and WLOG suppose we number these: $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$. Suppose P also contains a line, $\vec{x} + \lambda \vec{d}$.We have

$$\vec{a}_i \cdot \left(\vec{x} + \lambda \vec{d}\right) \geq b_i \; 1 \leq i \leq n$$

But this implies

$$\vec{a}_i \cdot \vec{x} + \lambda \vec{a}_i \cdot \vec{d} \geq b_i$$

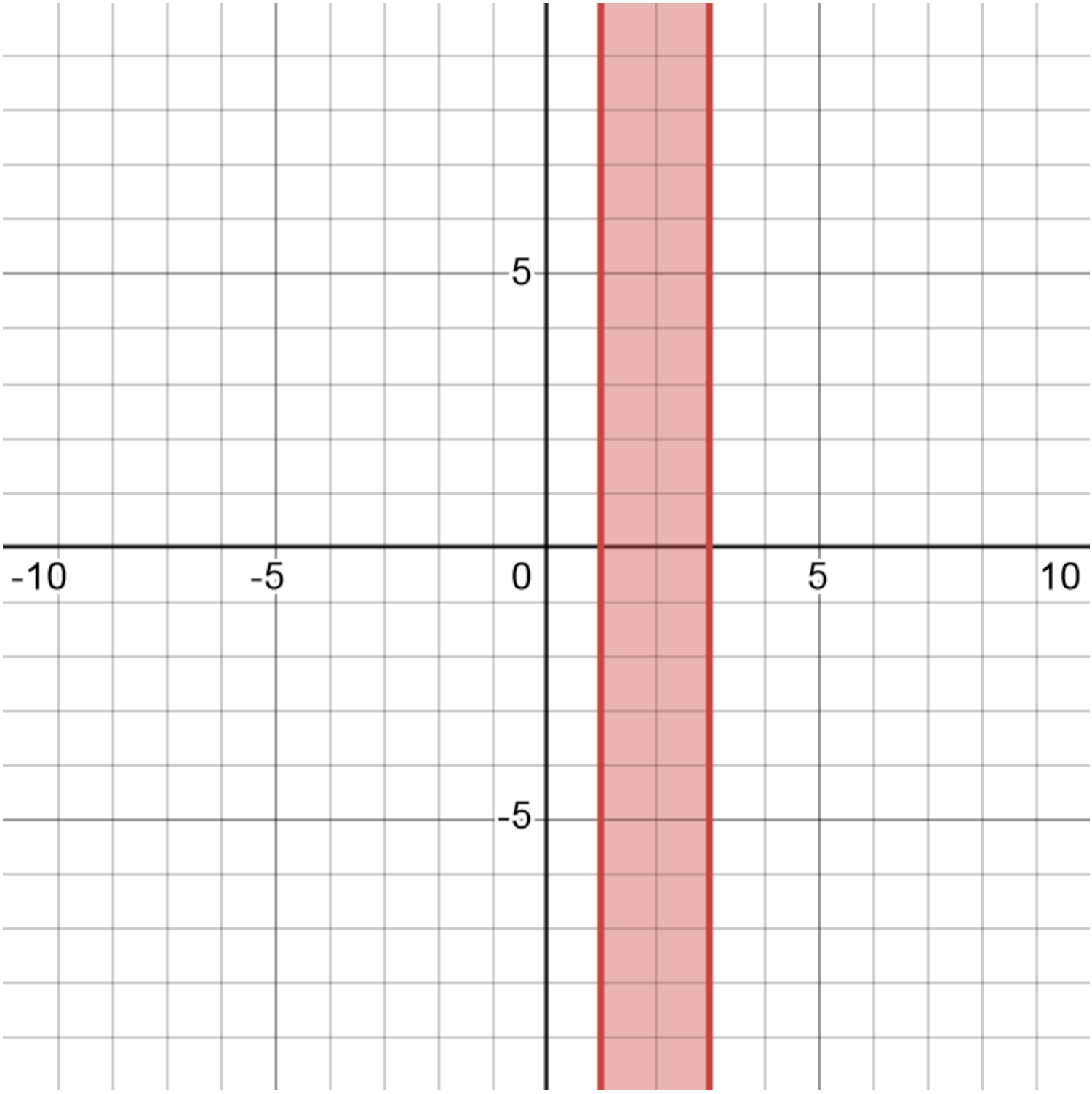
so we must have $\vec{d} = 0$ for $1 \leq i \leq n$. If not, we simply take λ large (and possibly negative) enough to violate the inequality for one of the i. If $\vec{d} = 0$, this contradicts the assumption that a line exists in P. Thus we cannot have a line in P and have n L.I. vectors simultaneously.

Corollary

Every non-empty polytope and every non-empty polyhedron in standard form has at least one BFS. This is because every polytope is line free. Every polyhedron in standard form is line free as well (see if you can prove this!)

Existence of Optimal Solutions

Question: Does no corner points imply an LP has no optimal solution? The answer is no! Recall out polyhedron from earlier:



If we take the objective function, $\min_{\vec{x} \in P} x$ then we see that there is an optimal solution. It isn't unique though. If an LP has not corner points, then it cannot have a unique optimal solution.

Theorem

Consider the L.P. $\min \{\vec{c} \cdot \vec{x} | \vec{x} \in P\}$ for some polyhedron P. If P has at least one extreme point and $\exists$ an optimal solution to the L.P., then $\exists$ an optimal solution which is an extreme point of P.

Proof

Let $v = \min\{\vec{c} \cdot \vec{x} | \vec{x} \in P\}$. Let Q be the set of all point where the optimal solution is achieved. If $P = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}\right\}$, then

$$Q = \left\{\vec{x} \in \mathbb{R}^n | A\vec{x} \geq \vec{b}, \vec{c} \cdot \vec{x} = v\right\}$$

Notice Q is still a polyhedron! Further, as P contains no lines and as $Q \subseteq P$, then Q also must contain no lines. This implies that there exists an extreme point in Q. Let one such point be $\vec{x}^*$. Now, we will show $\vec{x}^*$ must also be an extreme point of P. Suppose $\vec{x}^*$ isn't an extreme point in P. Then $\exists \vec{y}, \vec{z} \in P$ and $\lambda \in [0, 1]$ such that $\vec{y}, \vec{z} \neq \vec{x}^*$ and such that

$$\vec{x}^* = \lambda\vec{y} + (1 - \lambda)\,\vec{z}$$

As $\vec{x}^* \in Q$, then

$$v = \vec{c} \cdot \vec{x}^* = \vec{c} \cdot (\lambda\vec{y} + (1 - \lambda)\,\vec{z}) = \lambda\vec{c} \cdot \vec{y} + (1 - \lambda)\,\vec{c} \cdot \vec{z}$$

but as v is the minimum value of the objective function, we see

$$\vec{c} \cdot \vec{y}, \vec{c} \cdot \vec{z} \geq v$$

Thus

$$\lambda\vec{c} \cdot \vec{y} + (1 - \lambda)\,\vec{c} \cdot \vec{z} \geq \lambda v + (1 - \lambda)\,v = v$$

So $\vec{c} \cdot \vec{y} = \vec{c} \cdot \vec{z} = v$, which implies $\vec{y}, \vec{z} \in Q$. This would imply $\vec{x}^*$ is not an extreme point of Q, a contradiction! Thus $\vec{x}^*$ must be an extreme point of P as well. As $\vec{x}^*$ is in Q, this implies that $\vec{x}^*$ is an extreme point and optimal solution in P.

Corollary

Consider the L.P. $\min\{\vec{c} \cdot \vec{x} | \vec{x} \in P\}$ for some polyhedron P. Suppose P has at least one extreme point. Then either the limit of the optimal cost goes to $-\infty$ or $\exists$ an extreme point which is optimal. This implies that every non-empty polyhedron has an optimal cost that goes to negative infinity or there exists an optimal solution.

Remark

This is not the case for non-linear problems. Even for convex optimization over a polyhedron, this gets harder. Consider

$$\min_{x \geq 1} \frac{1}{x}$$

Notice that $0 < \frac{1}{x} \leq 1$ for $x \in [1, \infty)$. The optimal minimum doesn't exist, as we can get arbitrarily close to 0 without ever reaching 0. The limit of the objective doesn't go to $-\infty$ however.